

Business Requirements Document (BRD)

Hospital Management System Improvement Project

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1. Introduction

This Business Requirements Document (BRD) defines the detailed business needs for improving the Hospital Management System (HMS). The hospital is currently experiencing significant operational inefficiencies, including delayed patient processing, fragmented and outdated record-keeping systems, and weak interdepartmental communication among admissions, doctors, nurses, pharmacy, and administration.

The purpose of this project is to design and propose an integrated digital solution that improves operational efficiency, enhances the quality of patient care, and ensures secure, reliable access to medical information across all departments.

2. Business Problem Statement

The hospital faces the following core problems:

- Long patient waiting times due to manual and disconnected workflows
- Inefficient communication between departments, leading to delays in treatment
- Paper-based or outdated electronic record systems are causing data inconsistency
- Lack of a centralized patient information system
- Difficulty in tracking patient history and prescriptions in real time

These issues negatively impact patient satisfaction, staff productivity, and overall hospital performance.

3. Project Scope

3.1 In Scope

The project will focus on the following areas:

- Patient registration and admission management
- Electronic Medical Records (EMR) system
- Appointment scheduling and management
- Doctor consultation and treatment records
- Pharmacy prescription management and integration
- Internal communication between hospital departments
- Basic reporting and analytics dashboard for management

3.2 Out of Scope

The following are excluded from this phase:

- Insurance claims processing systems
- Advanced AI diagnostic tools

- External hospital network integrations
- Mobile app development (future phase consideration)

4. Stakeholder Identification and Analysis

Stakeholder	Role	Influence	Interest	Involvement
Hospital Administration	Decision making, funding approval	High	High	High
Doctors	Medical diagnosis and treatment	High	High	High
Nurses	Patient care and data updates	Medium	High	High
IT Department	System implementation and maintenance	High	High	High
Pharmacy Staff	Medication management and dispensing	Medium	High	Medium
Admissions Staff	Patient registration and data entry	Medium	High	High
Patients	End users receiving healthcare services	Low	High	Medium

5. Stakeholder Interview Questions (Elicitation)

For Doctors:

- What challenges do you face in accessing patient history?
- What improvements are needed in clinical documentation?

For Nurses:

- What issues occur during patient care coordination?
- How can patient record updates be improved?

For IT Department:

- What limitations exist in the current system infrastructure?
- What integration challenges are expected?

For Administration:

- What KPIs are most important for hospital performance?
- What reporting features are required for decision-making?

For Patients (feedback surveys):

- What difficulties do you face during registration or appointments?
- How can the hospital service experience be improved?

6. Project Objectives

- Reduce patient waiting time by improving workflow efficiency
- Improve inter-department communication and coordination
- Implement a centralized electronic medical record system
- Ensure real-time access to patient data across departments
- Enhance patient satisfaction and service quality
- Improve operational transparency and reporting capabilities

7. Functional Requirements

The system must support the following functions:

- Patient registration, search, and profile management
- Appointment booking and scheduling system
- Electronic Medical Records (create, update, retrieve)
- Doctor consultation notes and prescription management
- Pharmacy integration for medication tracking
- Inter-department communication module
- Patient history tracking and reporting
- Administrative dashboard for hospital performance metrics

8. Non-Functional Requirements

- **Security:** Role-based access control and encrypted patient data
- **Performance:** System response time under 3 seconds for record retrieval
- **Reliability:** 99% system uptime with minimal disruptions
- **Scalability:** Ability to handle increasing patient and data volume
- **Usability:** Simple and intuitive interface for all user groups
- **Compliance:** Must comply with healthcare data privacy regulations
- **Maintainability:** Easy system updates and bug fixes

9. Requirements Elicitation Techniques Used

- Structured interviews with key stakeholders
- Observation of current hospital workflows
- Workshops with cross-department teams
- Document analysis of existing medical records
- Surveys for patient experience feedback

10. System Process Overview (High-Level)

1. Patient arrives and registers through admissions
2. Patient data is stored in a centralized EMR system
3. The doctor accesses the patient's history and conducts a consultation
4. Prescription is generated and sent to the pharmacy system
5. Pharmacy dispenses medication and updates records
6. All updates are reflected in real-time across departments

11. Assumptions

- Staff will be trained on the new system
- Existing infrastructure can support system deployment
- Data migration from legacy systems is feasible

12. Constraints

- Limited project budget
- Fixed implementation timeline
- Resistance to change from staff
- Data quality issues in legacy systems

13. Risks

- User resistance to new system adoption
- Data migration errors from old systems
- System downtime during implementation
- Integration challenges between departments

14. Acceptance Criteria

- Reduced patient waiting time by at least 30%
- Successful integration of all core departments
- Real-time access to patient records without delays
- Positive user feedback from hospital staff
- System meets security and compliance standards

15. Conclusion

This BRD provides a comprehensive foundation for developing an improved Hospital Management System. It ensures that all stakeholder needs are captured, clearly defined, and aligned with the hospital's operational goals of efficiency, accuracy, and improved patient care delivery.

The End